ISHARA GLOVE · GRADUATION MVP · AMERICAN SIGN LANGUAGE

Sign Vocabulary v2.1 (ASL)

20 signs + rest + other
One glove · 5 flex sensors · MPU-6050
9 October 2026 · v2.1 · replaces v1

App version 1 uses **American Sign Language (ASL)**. Every sign below is the standard one-handed ASL form, and each one was picked so that **our hardware can tell it apart** from the others. The avatar teaches exactly these forms, so the glove must be recorded with exactly these forms.

One hand only
One glove, on the signing (dominant) hand. No sign may need the second hand.

No finger-to-finger touch sensing
A sign may include a touch (thumb on fingers, hand on chest), but its meaning must not depend on detecting it.

Different by bend, direction or motion
Each sign differs from the rest in which fingers bend (flex sensors) or how the hand points and moves (MPU-6050).

The 20 signs

#	WORD	LABEL	CATEGORY	HAND SHAPE (FLEX SENSORS)	WHERE · DIRECTION · MOTION (MPU-6050)	MAY BE CONFUSED WITH
1	**One**	`one`	NUMBERS	Index straight, others closed, thumb over them	Index up, palm toward you, still	Me, You
2	**Two**	`two`	NUMBERS	Index + middle straight (V), thumb holds ring + pinky	Fingers up, palm toward you, still	No
3	**Three**	`three`	NUMBERS	**Thumb + index + middle** straight (ASL form)	Fingers up, palm toward you, still	Water
4	**Four**	`four`	NUMBERS	4 fingers straight, thumb folded on the palm	Fingers up, palm toward you, still	Hello, Five
5	**Five**	`five`	NUMBERS	Whole hand open, fingers spread	Fingers up, still	Four, Bye
6	**Me**	`me`	PRONOUNS	Index straight	**Points to your own chest**, touches it	You, One
7	**You**	`you`	PRONOUNS	Index straight	**Points forward** at the other person	Me, One
8	**Hello**	`hello`	SOCIAL	Flat hand (B), fingers together, thumb folded	**Salute:** fingertips at the side of the forehead, hand moves out and away	Four, Thank you
9	**Thank you**	`thanks`	SOCIAL	Flat hand (B)	Fingertips at the chin, hand moves forward and down, ends palm up	Please, Hello
10	**Please**	`please`	SOCIAL	Flat hand (B), fingers together	Palm flat on the chest, circles	Sorry, Thank you
11	**Yes**	`yes`	SOCIAL	Fist (S), thumb across the fingers	Wrist nods up and down, like a head nodding	Sorry
12	**No**	`no`	SOCIAL	Index + middle straight together, **snap shut onto the thumb**	In front of the shoulder, snaps twice	Two
13	**Sorry**	`sorry`	SOCIAL	Fist (A), thumb straight up along the side	On the chest, circles	Yes, Please
14	**Water** NEW	`water`	NEEDS	**Index + middle + ring** straight (W), thumb holds the pinky	Index fingertip taps the chin twice	Three
15	**Bye** NEW	`bye`	SOCIAL	Open hand, **fingers bend down and up together**, twice or more	Palm out, at shoulder height, the arm stays still	Five, Hello
16	**Come**	`come`	ACTIONS	Index **curls in and straightens** (one-handed form)	Palm up, beckons toward you	One, You
17	**Drink**	`drink`	NEEDS	C shape, as if holding a cup	At the mouth, tilts toward you	Hungry
18	**Call / Phone**	`call`	ACTIONS	Thumb + pinky straight (Y)	Beside the ear and cheek	I love you
19	**I love you**	`iloveyou`	FEELINGS	Thumb + index + pinky straight (ILY)	Palm out, **held still** in front (no push)	Call
20	**Hungry**	`hungry`	NEEDS	C shape	On the upper chest, moves down once	Drink
+	**No sign**	`rest`	EXTRA CLASS	Hand relaxed	Hand still or resting, not signing	—
+	**Everyday movement**	`other`	EXTRA CLASS	Any	Typing, phone, cup, scratching, talking with the hands	—

Numbers 1–5 are signed with the palm toward you (ASL counting). Yes uses the S fist (thumb across the fingers); Sorry uses the A fist (thumb up along the side). The four finger sensors read both as a fist; only the thumb sensor differs, so the main difference the glove sees is the motion (nod vs. circle on the chest).

What changed from v1

- **Good and Bad were removed.** Thumb up / thumb down is a gesture, not the ASL sign. ASL GOOD and BAD both start with a flat hand at the chin, almost the same as THANK YOU, so one glove can't tell them apart. They are replaced by **Water** (`water`) and **Bye** (`bye`).
- **Hello** is now the ASL salute from the forehead, not a side-to-side wave.
- **No** is now the ASL snap (index + middle onto the thumb), not an index finger shaking.
- **Three** is fixed to the ASL form: thumb + index + middle.
- New extra class `other` for everyday movements (AI v3).
- **v2.1:** Please is a flat B hand; Sorry is the A fist; I love you is held still. No label changed.

Check before recording

1. **Watch each sign in an ASL video dictionary** (for example Lifeprint / ASL University or Handspeak) and film one team member doing it as our reference. Everyone records from that reference.
2. **Don't mix Three and Water.** Three = thumb + index + middle. Water = index + middle + ring, with the thumb holding the pinky.
3. **The flat-hand group** (Four, Five, Hello, Thank you, Please, Bye) shares almost the same bend and differs by place and motion. If the model keeps mixing them after training, drop Please first.
4. **Come** uses the one-handed form (only the glove hand beckons).
5. **The rest and other classes are essential.** Without them the model outputs words while the hand is doing nothing or doing everyday things.

Demo sentences

SENTENCE	SIGNS (ASL ORDER)
I am hungry	`me` + `hungry`
I want water	`me` + `water`
Water, please	`water` + `please`
You, come	`you` + `come`
Call me	`call` + `me`
Hello, thank you	`hello` + `thanks`
Sorry, bye	`sorry` + `bye`
I love you	`iloveyou`

Best demo for the committee

Same or almost the same hand shape: only direction, place or motion differs, which proves the motion sensor is doing real work.

PAIR	DIFFERENCE
Me / You	Same finger, toward yourself or forward
Yes / Sorry	Fist (S / A), nod or circle on the chest
Thank you / Please	Same flat hand, from the chin forward or circling on the chest
Three / Water	Three fingers each, but different fingers (flex sensors)

Backup signs (ASL, one-handed)

WORD	LABEL	WHAT MAKES IT DIFFERENT
Like	`like`	Only the middle finger bends; pulls away from the chest
Where	`where`	Index up, shakes side to side
Mother	`mother`	Open hand, thumb taps the chin
Home	`home`	Fingertips together, mouth then cheek

Recording plan

- **22 classes** (20 signs + `rest` + `other`) × **20 takes each** × **at least 5 people** ≈ 2,200 takes, about 75 minutes of recording. Spread it over several days.
- Calibrate each person first: `python record.py calib ahmed`
- Then each sign: `python record.py water ahmed` , and everyday movements: `python record.py other ahmed`
- Long wear sessions with no signing: `python record.py wear ahmed 60` . They teach the model to stay quiet.
- **Test accuracy on a person who was not recorded.** That is the number to present at the defense.
- Review the confusion matrix after each training run, and replace signs that keep getting mixed up with a backup sign.